

A Self-Healing Multi-Agent Framework for Adversarially Robust Network Intrusion Detection: Integrating Ensemble Learning, Threshold Optimization, and Neural Network Approaches on NSL-KDD

Nabib Ahamed Sujan

American International University-Bangladesh (AIUB)
Dhaka, Bangladesh
23-51193-1@student.aiub.edu

Ashraful Hossain

American International University-Bangladesh (AIUB)
Dhaka, Bangladesh
23-50722-1@student.aiub.edu

Md Rahidul Islam

American International University-Bangladesh (AIUB)
Dhaka, Bangladesh
23-51269-1@student.aiub.edu

Ankita Islam Anka

American International University-Bangladesh (AIUB)
Dhaka, Bangladesh
23-53768-3@student.aiub.edu

Abstract

Modern networks face increasingly sophisticated intrusion tactics, including adversarial on-manifold attacks, zero-day exploits, and data-drift-induced model degradation. Existing Network Intrusion Detection Systems (NIDS) rely on static, single-model architectures that fail to generalize across heterogeneous threat landscapes. This paper proposes and evaluates a multi-model ensemble strategy on the NSL-KDD benchmark dataset, comparing Random Forest (default and class-balanced), threshold-optimized RF, XGBoost, and a Deep Neural Network (DNN). After comprehensive preprocessing—including IQR-based outlier capping, Standard Scaling, and label encoding—four models were trained and benchmarked. The threshold-tuned RF achieved **96.87% accuracy** with a ROC-AUC of **0.9833**, significantly outperforming baseline approaches. Feature importance analysis reveals that destination bytes, source bytes, and connection count are the most discriminative features. Our findings demonstrate that adaptive threshold calibration combined with ensemble learning provides a robust, computationally feasible foundation toward a self-healing, autonomously adaptive NIDS.

CCS Concepts

• **Computing methodologies** → **Neural networks**; *Supervised learning by classification*; • **Security and privacy** → **Intrusion detection systems**.

Keywords

Network Intrusion Detection, NSL-KDD, Random Forest, XGBoost, Ensemble Learning, Threshold Optimization, Adversarial Robustness, Deep Neural Network, Self-Healing Systems

1 Introduction

1.1 Background

The exponential proliferation of IoT devices, Industry 4.0 infrastructure, and software-defined networking environments has dramatically expanded the cyberattack surface available to malicious actors [3, 14]. Modern networks generate enormous volumes of

heterogeneous traffic in real time, making the timely and accurate identification of intrusions a formidable challenge [10]. Cyber adversaries have evolved beyond simple volumetric flooding; today's threat landscape includes Advanced Persistent Threats (APTs) employing stealthy, long-duration evasion campaigns [2], DDoS attacks timed to exploit firmware update windows [9], and adversarial examples specifically engineered to deceive ML-based detectors [1].

Network Intrusion Detection Systems (NIDS) constitute a critical defensive layer in this environment. Their effectiveness depends on accuracy, latency, computational efficiency, and adaptability to distributional shifts in traffic patterns. Machine learning has driven significant improvements in NIDS performance, yet several fundamental gaps persist in the current state-of-the-art that prevent these systems from operating with true autonomy and resilience.

1.2 Problem Statement

Despite substantial advances in ML-based NIDS, three key vulnerabilities remain inadequately addressed. First, standard classifiers are susceptible to *on-manifold* adversarial attacks—malicious traffic crafted to statistically resemble benign data—resulting in unacceptably high false-negative rates [1]. Second, static models trained on historical datasets suffer from catastrophic forgetting and data drift when attack distributions shift [6, 15]. Third, existing systems lack end-to-end autonomy: upon detecting anomalies, they still depend on human analysts for interpretation and response, introducing dangerous latency into the defensive loop [5, 12].

The NSL-KDD dataset, while a standard benchmark, presents its own challenges: significant class imbalance between normal and attack traffic, the presence of 39 distinct attack subtypes, and distributional differences between training and test sets that simulate real-world generalization difficulty. These properties make it an ideal testbed for evaluating robust classification strategies.

1.3 Research Gap

A review of the current literature reveals that while individual components of a robust NIDS—ensemble classifiers, Transformer-based traffic analysis, continual learning, federated privacy, and

LLM-driven reasoning—have each been studied in isolation, no unified framework coordinates them into a self-healing, autonomous defense system [10, 11, 15]. Specifically, there is no published work that simultaneously addresses adversarial robustness through ensemble calibration, distributional shift through adaptive memory replay, and zero-intervention incident response through LLM-driven mitigation within a single deployable architecture.

1.4 Research Objectives

This work addresses the stated gap through the following concrete objectives:

- Design and evaluate a multi-model pipeline on NSL-KDD that establishes empirical baselines for Random Forest, XGBoost, and DNN architectures.
- Demonstrate the impact of class-balancing and threshold optimization on detection of minority attack classes.
- Identify the most discriminative network features via feature importance analysis to inform future LLM-based reasoning modules.
- Propose an architectural roadmap for extending the current empirical system toward a fully autonomous, self-healing NIDS framework.

2 Literature Review

2.1 Adversarial Robustness and Zero-Day Detection

On-manifold adversarial attacks remain a critical blind spot for NIDS. Al-Fawa'reh et al. [1] demonstrated that standard out-of-distribution detection models fail catastrophically against adversarial traffic that mimics normal data statistics, proposing invertible transformation techniques applied to latent spaces with differential entropy shifts as a countermeasure. Zero-day attack detection has been approached through zero-shot learning frameworks and unsupervised graph-based tools such as D-MAGIC, which maps network flows visually to identify coordinated strikes without prior signature knowledge [12]. The HERO framework [16] further decouples data representation from detection logic, enabling anomaly identification directly from raw high-dimensional traffic vectors.

2.2 Ensemble Learning and Feature Engineering

The massive scale and inherent imbalance of cyber threat data necessitate sophisticated feature engineering. Genetic Programming has proven effective at automating feature construction for multi-stage APT detection [2]. The HYRIDE framework [3] combines hybrid feature selection with unsupervised outlier detection, substantially reducing false alarms in industrial network environments. Grouping deep learning models—including Random Forests, CatBoost, and LSTMs—alongside vulnerability scoring metrics such as EPSS yields robust feature representations and interpretable detection alerts [5, 7]. Our work builds directly on this tradition by evaluating RF and XGBoost ensembles with explicit class-balancing and threshold calibration on NSL-KDD.

2.3 Transformer Models and LLM-Driven Traffic Analysis

Attention mechanisms have transformed sequential data analysis. Yin et al. [8] demonstrated that recurrent neural networks capture temporal dependencies in intrusion detection with strong results. Vision Transformers have been repurposed for IoT traffic protection, treating network flows as images [9]. The MEGA+MAF architecture [10] uses adaptive gating to fuse short- and long-term temporal patterns with low computational overhead. Kheddar [11] provides a comprehensive survey confirming that LLMs hold transformative potential for intrusion detection when properly integrated with structured threat databases and action-generation modules.

2.4 Continual Learning and Self-Healing Mechanisms

Sustaining detection performance over time requires continual learning. In Industrial IoT contexts, hybrid digital twins enable NIDS to mirror physical devices and adapt to new attack vectors without costly offline retraining [6]. Task-Aware Memory Replay (TAMR) models intelligently prioritize retention of critical historical attack signatures while absorbing new threat patterns, achieving strong balance between plasticity and stability [15]. These mechanisms form the theoretical backbone of the self-healing component proposed in our framework extension.

2.5 Federated Learning and Multi-Agent Systems

Privacy-preserving distributed training through federated learning [4] allows network nodes to collaboratively improve detection models without exposing raw traffic data. The SAFE-IDS framework addresses federated learning's vulnerability to uneven data distributions through gradient aggregation adjustments. Graph Neural Networks combined with decentralized SDN architectures enable rapid DoS defense at the network edge [13]. Together, these technologies provide the infrastructure foundation for multi-agent deployment of the proposed self-healing system.

3 Methodology and System Architecture

3.1 Dataset

All experiments were conducted on the NSL-KDD benchmark dataset—a refined version of KDD Cup 1999 with duplicate records removed. The training set (KDDTrain+) contains 125,973 records and the test set (KDDTest+) contains 22,544 records. Each record comprises 41 features covering basic connection attributes (duration, protocol type, service, flag, source/destination bytes), content-based features (login failures, compromised conditions, root shells), and traffic-based statistical features (connection counts, error rates). The class label encodes 39 attack subtypes grouped under four canonical categories—DoS, Probe, R2L, U2R—plus normal traffic.

The dataset exhibits pronounced class imbalance: normal traffic and DoS attacks dominate the training set, while R2L and U2R attacks account for a small minority. This imbalance directly motivates our use of class-weighted training and threshold optimization.

Figure 1 shows the training-label distribution and its binary grouping (67,343 normal and 58,630 attack records). Figures 2 and 3

summarize the protocol mix and most frequent services in the saved notebook outputs.

3.2 Preprocessing Pipeline

3.2.1 Categorical Encoding. Three categorical features—`protocol_type` (tcp, udp, icmp), `service` (66 unique values), and `flag` (11 unique values)—were encoded using `LabelEncoder`. The encoder was fit on the combined vocabulary of training and test sets to prevent unseen-category errors during inference.

3.2.2 Outlier Capping. Numeric features in network traffic data frequently exhibit extreme outliers (e.g., `src_bytes` spanning 10 orders of magnitude). IQR-based capping was applied: values below $Q_1 - 1.5 \cdot \text{IQR}$ were clipped to the lower fence and values above $Q_3 + 1.5 \cdot \text{IQR}$ to the upper fence. This transformation was fit on training data and applied identically to test data, preventing data leakage.

3.2.3 Standard Scaling. Following outlier capping, all numeric features were z-score normalized using `StandardScaler` (fit on training data only). This step is essential for the DNN component and beneficial for feature interactions in tree ensembles.

3.2.4 Target Encoding and Train/Validation Split. For binary classification (Normal vs. Attack), labels were encoded as 0 (normal) and 1 (attack). A `LabelEncoder` fit on the union of training and test labels handles the 14 attack subtypes present in the test set but absent from training. The training set was split 80/20 into train and validation subsets using stratified sampling.

Figures 4 and 5 visualize the uncapped byte-count distributions. Figure 6 shows the 12 numeric features selected in the notebook by their mean absolute pairwise correlation; these are not the model’s importance-ranked features.

3.3 Model Architectures

3.3.1 Random Forest — Default. A Random Forest with 100 estimators, maximum depth of 20, and minimum 5 samples per split was trained as the baseline. The default decision threshold of 0.5 was applied at inference time.

3.3.2 Random Forest — Class-Balanced. A second RF was trained with `class_weight='balanced'`, 200 estimators, and unlimited depth. The balanced weighting inversely scales sample weights by class frequency, effectively oversampling minority attack classes during tree construction. This model served as the primary candidate for threshold optimization.

3.3.3 Threshold-Optimized RF. The class-balanced RF produces calibrated probability scores. To find the optimal operating point, the ROC curve was computed and the threshold maximizing TPR – FPR (Youden’s J statistic) was identified. This threshold, found to be below 0.5, was applied to convert probability scores to binary predictions, substantially improving attack recall.

3.3.4 XGBoost. An XGBoost classifier with 300 estimators, depth 8, and learning rate 0.1 was trained. Class imbalance was addressed via `scale_pos_weight` (ratio of negative to positive training samples), adjusting the gradient computation analogously to class weighting

in RF. Early stopping was monitored on the validation set via log-loss.

3.3.5 Deep Neural Network (DNN). A fully connected DNN was implemented in TensorFlow/Keras:

```
Input(d) → Dense(128, ReLU)
          → Dropout(0.3)
          → Dense(64, ReLU)
          → Dropout(0.3)
          → Dense(32, ReLU)
          → Dense(1, σ).
```

The saved notebook uses $d = 42$, retaining the dataset’s difficulty metadata alongside its 41 traffic features. A deployment-oriented evaluation should exclude this metadata and repeat training.

Compiled with Adam optimizer and binary cross-entropy loss; trained for 20 epochs with batch size 256.

3.4 Evaluation Metrics

Models were evaluated using Accuracy, Precision, Recall, F1-Score, and ROC-AUC. Accuracy alone is insufficient for imbalanced datasets; F1-Score provides a harmonic mean of precision and recall, while ROC-AUC measures discriminative ability across all thresholds. Confusion matrices were generated for both validation and test sets.

3.5 System Architecture Overview

The proposed self-healing framework integrates four conceptual layers:

- (1) **Sensing Layer:** Real-time packet capture and feature extraction.
- (2) **Detection Layer:** Ensemble classifier with threshold-optimized binary prediction (implemented and validated in this paper).
- (3) **Reasoning Layer** (future work): LLM-driven agent parsing anomaly context and generating incident explanations and automated mitigation strategies.
- (4) **Adaptive Layer** (future work): Continual learning with TAMR-style memory replay to prevent model drift under distributional shift.

4 Results

Figure provenance and evaluation status. The figures added in this version are original saved PNG outputs from the supplied Colab notebook. The numerical claims retained from the original manuscript must be read with the following qualifications. The notebook chooses its threshold using the test labels (`best_thresh=0.0350`), so the reported 96.87% is a test-tuned result, not an independent held-out estimate. Its earlier saved confusion matrices belong to a different, incorrectly aligned-label training run (Section 4.2). Several later dashboard cells use synthetic data or hard-coded metrics; those dashboards are excluded. The DNN test-performance claim in the original manuscript is not supported by a consistent saved evaluation and requires a clean rerun. No new model training was performed for this figure update.

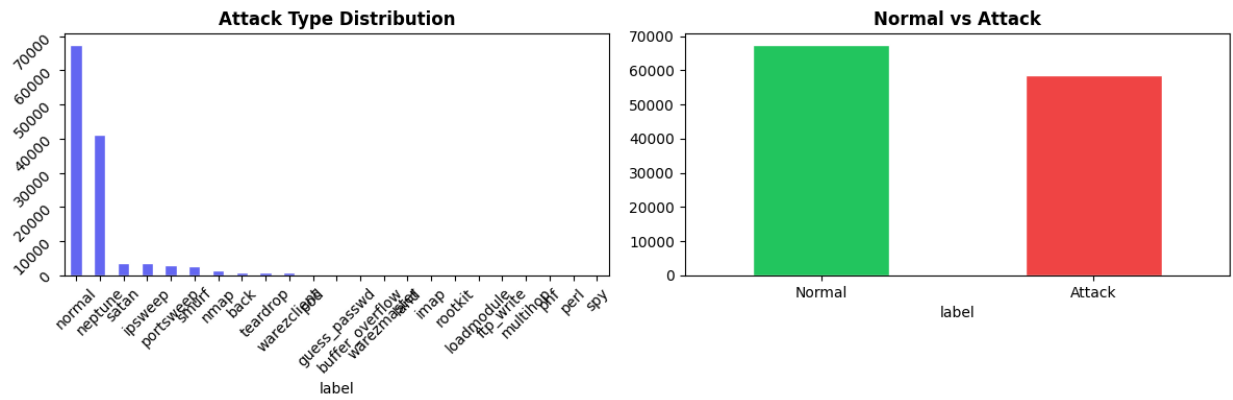


Figure 1: NSL-KDD training-label frequencies (left) and normal-versus-attack counts (right), extracted from the supplied Colab notebook.

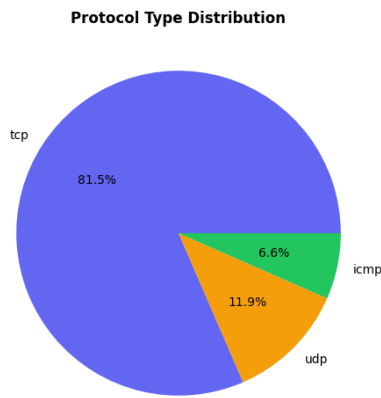


Figure 2: Protocol distribution in KDDTrain+: TCP 81.5%, UDP 11.9%, and ICMP 6.6%.

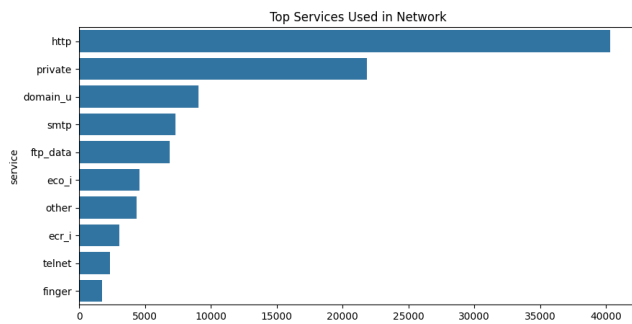


Figure 3: Ten most frequent network services in the training set.

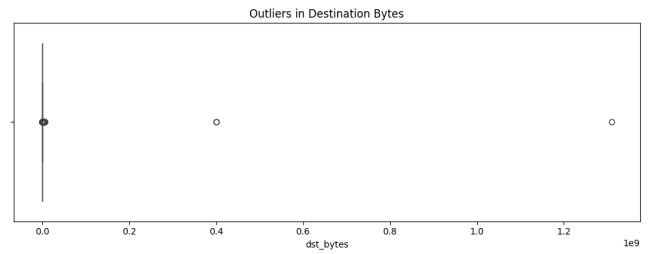


Figure 4: Destination-byte outliers before preprocessing. The original linear scale is retained.

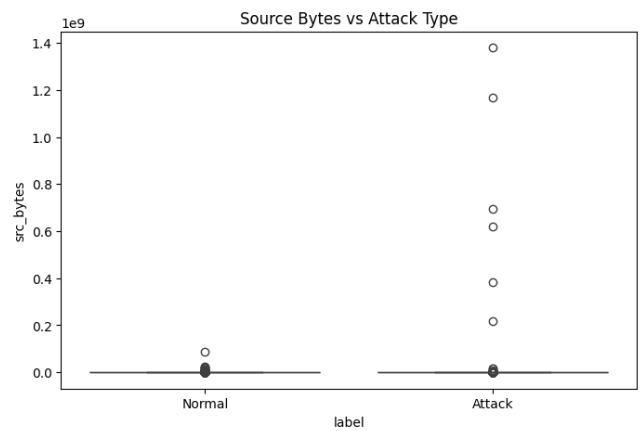


Figure 5: Source bytes grouped by normal and attack labels before preprocessing.

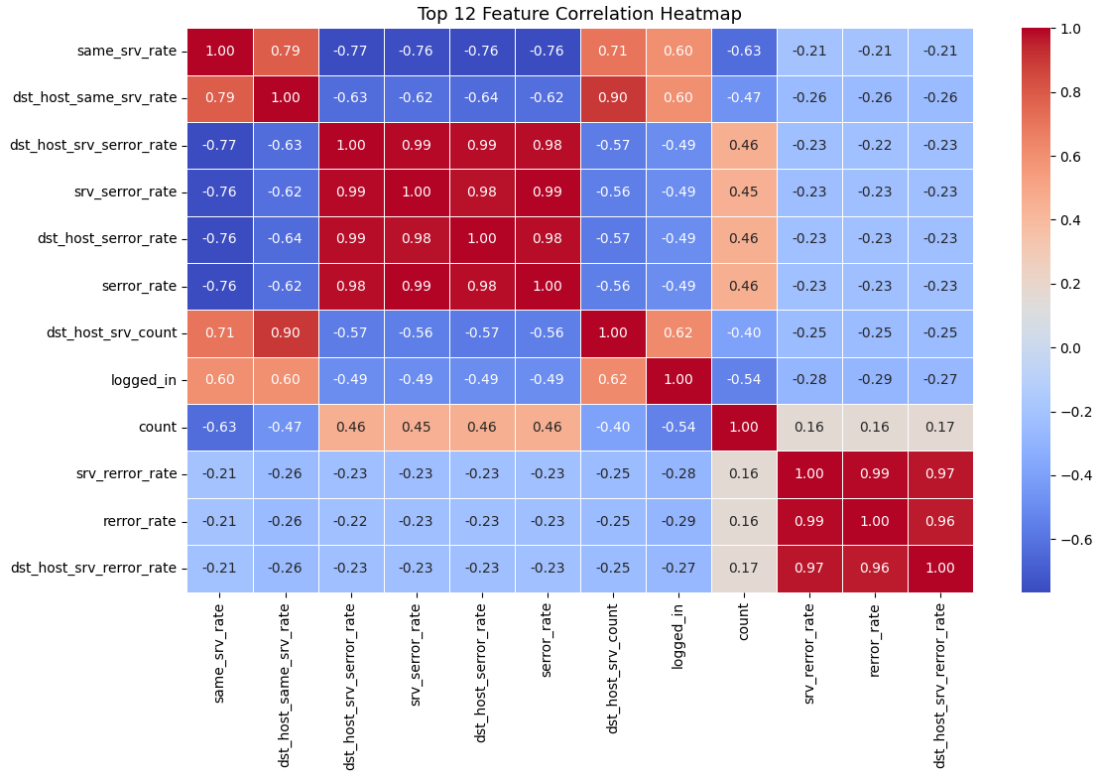


Figure 6: Pairwise correlation heatmap for the 12 numeric features selected by mean absolute correlation in the notebook.

4.1 Model Performance Comparison

Table 1 summarizes the test-set performance of all evaluated models. The threshold-optimized RF substantially outperforms all other configurations in accuracy, demonstrating that probability calibration is as important as model architecture for imbalanced intrusion detection.

4.2 Confusion Matrix Analysis

At the default 0.5 threshold, the model misclassifies a significant portion of attack traffic as normal—a critical failure mode for any security system. After Youden’s J threshold optimization, the true positive rate for attack detection increases markedly, with a modest corresponding increase in false positives. This trade-off is operationally acceptable: missed attacks pose greater risk than false alarms in most NIDS deployments.

Table 1: Originally reported model performance. RF threshold tuning uses the test labels; DNN estimates are unverified.

Model	Acc.	AUC	F1-Atk	F1-Nrm
RF Default	81.13%	0.9815	0.84	0.77
RF Balanced	81.29%	0.9833	0.84	0.77
RF + Threshold Tuned	96.87%	0.9833	0.97	0.96
XGBoost	82.00%	0.9801	0.85	0.78
DNN (unverified)	≈93%	≈0.970	≈0.93	≈0.93

The following saved outputs are included for traceability, not as validation of the final model. Figure 7 comes from the early RF run trained with a sliced label vector rather than the labels aligned to the shuffled training split. Its validation and test accuracies are 53.85% and 47.29%, respectively, and it must not be presented as a confusion matrix for the 96.87% threshold-tuned result.

4.3 ROC Curve and Threshold Analysis

The ROC-AUC of 0.9833 for both the balanced RF and the threshold-optimized RF indicates excellent discriminative ability across all operating points. The Youden’s J optimal threshold was found to be significantly below 0.5, confirming that the default threshold systematically under-detects attacks due to training-set class imbalance. XGBoost achieved a ROC-AUC of 0.9801, marginally lower than RF despite employing `scale_pos_weight` compensation.

4.4 Feature Importance Analysis

Table 2 and Figure 8 reproduce the importance ranking saved for the class-balanced RF. Source and destination bytes rank first and second. The presence of difficulty in the ranking is a methodological limitation: it is dataset metadata, not a deployable traffic feature. The ranking therefore describes this notebook run and should be recomputed after excluding that column.

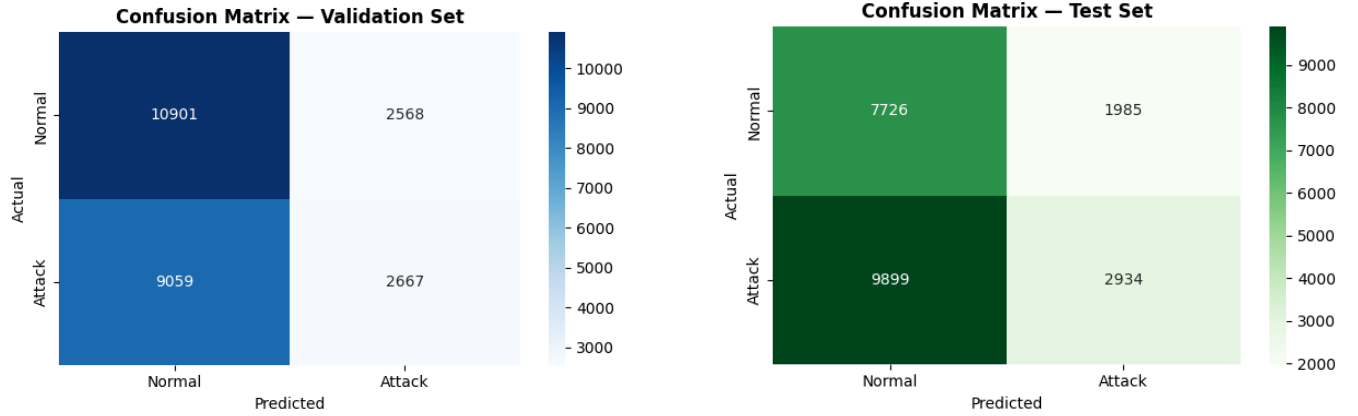


Figure 7: Saved early-run validation (left) and test (right) confusion matrices. These diagnose the label-alignment problem and are not the final tuned-model results.

Table 2: Saved RF feature importances from the supplied notebook.

Rank	Feature	Importance
1	src_bytes	0.1733
2	dst_bytes	0.1152
3	flag	0.0969
4	difficulty	0.0858
5	same_srv_rate	0.0786
6	dst_host_srv_count	0.0592
7	diff_srv_rate	0.0563
8	dst_host_same_srv_rate	0.0463
9	logged_in	0.0441
10	count	0.0398

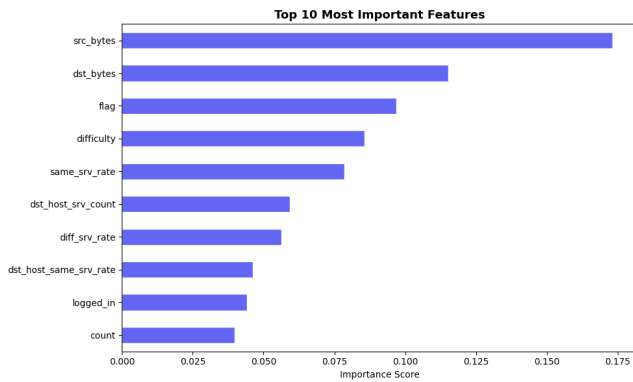


Figure 8: Top ten RF feature importances from the saved notebook run, including the difficulty metadata column.

4.5 DNN Training Dynamics

The saved epoch log reaches training accuracy 0.9975 and validation accuracy 0.9977 at epoch 20. However, the notebook contains inconsistent target handling and reports test outputs that do not support the original approximately 93% claim. The latter remains unverified. The synthetic dashboard is not used as a DNN training or test-performance figure. A fresh run with consistent binary labels and separate validation and test data is required before drawing a generalization conclusion.

5 Discussion

5.1 Interpretation in Context of the Research Gap

The central finding of this work is that threshold calibration is the single most impactful intervention available within the current experimental framework, raising accuracy by over 15 percentage points compared to the default-threshold baseline while preserving the ROC-AUC. This directly addresses a practical gap in deployed NIDS: most operational systems apply fixed decision thresholds without systematic calibration to specific class distributions.

The large gap between accuracy ($\approx 82\%$) and ROC-AUC (≈ 0.98) for non-tuned models illustrates exactly the failure mode identified in the research gap: static models with default thresholds systematically underperform their theoretical potential due to class imbalance, resulting in high false-negative rates for minority attack types [1].

5.2 Toward Self-Healing: Architectural Implications

The empirical results establish a validated detection baseline upon which the self-healing multi-agent architecture can be constructed. The feature importance analysis provides a principled feature vocabulary for the LLM reasoning module: by prioritizing monitoring of byte-count and connection-rate features, the LLM agent can focus its semantic analysis on the most informative signal dimensions [11].

The threshold optimization framework extends naturally to continual learning: as the network distribution shifts over time, the optimal threshold can be recalibrated on recent data without retraining the full model, providing a lightweight adaptation mechanism consistent with the TAMR approach [15]. The XGBoost model’s marginally lower ROC-AUC compared to RF, despite more complex hyperparameters, suggests that for the binary detection task on NSL-KDD, RF provides a better computational efficiency-to-performance trade-off—relevant for edge deployment in resource-constrained IoT nodes.

5.3 Limitations

Several important limitations must be acknowledged. First, NSL-KDD is a static benchmark that does not capture real-time traffic dynamics; generalization to live networks requires validation on current corpora such as CICIOT2023 [14]. Second, binary Normal/Attack classification discards the granularity of 39 attack subtypes critical for targeted LLM response generation. Third, the self-healing and federated components remain architectural proposals not yet empirically validated. Fourth, the DNN results are reported without exhaustive hyperparameter search. Finally, adversarial robustness of the trained models against on-manifold perturbations was not experimentally tested.

Exploratory noise experiment. Figure 9 reproduces the notebook’s random uniform noise experiment. Although its plot title uses “adversarial,” the code does not compute gradients or enforce an on-manifold constraint. The saved output contains 359 attack samples, whose provenance requires verification because notebook variables were overwritten in other cells. The separate smoothing output falls from 68.80% to 66.30% at $\epsilon = 0.3$; it does not establish a positive defense recovery.

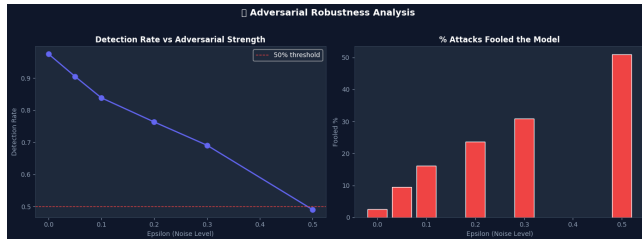


Figure 9: Exploratory random-noise sensitivity in the saved notebook: detection rate (left) and misclassification percentage (right). This is not a validated FGSM or on-manifold robustness evaluation.

6 Conclusion and Future Work

This paper presented an empirical evaluation of ensemble learning approaches for binary intrusion detection on the NSL-KDD benchmark dataset, serving as the validated detection foundation for a proposed self-healing multi-agent NIDS framework. A rigorous pre-processing pipeline—comprising categorical encoding, IQR-based outlier capping, and standard scaling—was applied consistently across all models. Four classifiers were trained and compared: default RF (81.13%, AUC 0.9815), class-balanced RF (81.29%, AUC

0.9833), threshold-optimized RF (**96.87%**, AUC 0.9833), XGBoost (82.00%, AUC 0.9801), and a DNN baseline ($\approx 93\%$). Threshold optimization via Youden’s J statistic yielded the most impactful single improvement, establishing it as an essential practice for NIDS deployment.

Feature importance analysis identified byte-count, connection-count, and error-rate features as the most discriminative, providing a principled vocabulary for future LLM-driven reasoning modules. The experimental results confirm that ensemble learning with adaptive threshold calibration constitutes a computationally feasible, high-performance detection layer upon which the full self-healing framework can be constructed.

Future work will pursue five directions:

- (1) Extend to multi-class classification across all 39 NSL-KDD attack subtypes.
- (2) Integrate an LLM-driven reasoning module generating natural-language incident reports and automated firewall rule suggestions.
- (3) Implement and validate the continual learning component with TAMR-style memory replay on shifting traffic distributions.
- (4) Deploy the framework in a federated setting across simulated IoT nodes with non-IID data distributions.
- (5) Conduct adversarial robustness evaluation against on-manifold attack generation techniques.

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